

Generation Interconnection System Impact Study Report

For Queue Project Q17-0853

Cement Plant 138 kV

April 02, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0853. The Interconnection Customer proposes a 75.9 MW battery energy storage facility interconnecting at the Cement Plant 138 kV substation in the Virginia area. The request was received on August 26, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 311 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,941,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Pocket / Mt Vernon 138.0 kV	N Pt 345-138 kV Autotransformer	115.4	291	9.7
North Pt Substation 345 kV/138 kV TX	Dewey / Dowell 138.0 kV	114.9	362	12.0
Pocket / Tillamook 138.0 kV	N Pt 345-138 kV Transformer	114.7	302	9.7
Tillamook / Dewey 138.0 kV	N Pt 345/138kV Transformer Bank	114.1	311	9.7
Dewey / Dowell 138.0 kV	N Point 345-138 kV Auto	112.1	320	9.7
West Paintsville / Dowell Substation 138.0 kV Line	Wissota / Nsp 138.0 kV L	107.8	427	11.6
Beardsley St / Red Hills 138.0 kV	Lowmoor / Hinton Ap 345.0 kV	107.4	427	8.0
Lomoor 345/138kV Autotransformer	Aspen / Lowmoor 345.0 kV	105.0	449	8.5
Mt Vernon / Kettle Crk Primary 138.0 kV	N Point 345/138 kV Autotransformer	102.8	319	9.7
Neptune / Kettle Crk Pimary 138.0 kV L	N Pt 345/138kV XFMR	102.7	307	9.7
Orovada / Carlton Hills 138.0 kV	Clinch Riv / Carlton Hills 345.0 kV	100.8	282	8.1
Mcclain Enrgy Facility / Coal 138.0 kV	Manatee / Hinton Ap 345.0 kV line	100.8	323	5.6
North Sweetwater Substation / Scarbro 138.0 kV	Phelan Substation / Bradley Substation 138.0 kV	100.5	577	5.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Pocket / Mt Vernon 138.0 kV	Rebuild line with higher-capacity conductor	\$8,367,000	6% (\$505,000)
N Pt 345/138 kV XFMR	Replace transformer with higher-rated unit	\$17,716,000	4% (\$612,000)
Pocket / Tillamook 138.0 kV	Rebuild line with higher-capacity conductor	\$11,052,000	6% (\$667,000)
Tillamook Substation / Dewey 138.0 kV Line	Rebuild line with higher-capacity conductor	\$9,634,000	6% (\$582,000)
Dewey / Dowell 138.0 kV	Rebuild line with higher-capacity conductor	\$6,161,000	6% (\$372,000)
Dowell / West Paintsville 138.0 kV	Reconductor line	\$1,301,000	4% (\$46,000)
Beardsley St Substation / Red Hills 138.0 kV	Reconductor line	\$5,544,000	2% (\$118,000)
Lowmoor 345/138kV XFMR	Replace transformer with higher-rated unit	\$18,260,000	2% (\$319,000)
Mt Vernon / Kettle Crk Primary 138.0 kV	Reconductor line	\$977,000	6% (\$59,000)
Neptune / Kettle Crk Primary 138.0 kV L	Reconductor line	\$7,184,000	6% (\$434,000)
Carlton Hills / Orovada 138.0 kV	Reconductor line	\$155,000	6% (\$9,000)
Mcclain Enrgy Facility / Coal 138.0 kV	Reconductor line	\$3,255,000	6% (\$196,000)
North Sweetwater Substation / Scarbro 138.0 kV	Reconductor line	\$1,444,000	2% (\$22,000)
	Total allocated to project		\$3,941,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.